

SQL Advance Level Question and Answers:

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Question 1] What are triggers and how are they used?

Answer] A **trigger** is a database object that automatically executes a predefined action when a specific event (INSERT, UPDATE, DELETE) occurs on a table or view.

Triggers are used for auditing, enforcing business rules, and maintaining data integrity.

Example:

```
CREATE TRIGGER before_insert_log
BEFORE INSERT ON orders
FOR EACH ROW
INSERT INTO audit_log VALUES (NEW.order_id, NOW());
```

Question 2] Explain the concept of ACID in databases.

Answer] ACID ensures reliable transactions:

- **Atomicity** – Transaction is all or nothing
 - **Consistency** – Database remains valid after transaction
 - **Isolation** – Concurrent transactions don't affect each other
 - **Durability** – Committed data is permanently saved
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Question 3] What is a stored procedure?

Answer] A **stored procedure** is a reusable block of SQL code stored in the database. It improves performance, security, and reduces code duplication.

Example:

```
CREATE PROCEDURE get_users()
BEGIN
    SELECT * FROM users;
END;
```

Question 4] How do you manage transactions in SQL?

Answer] Transactions are controlled using:

- BEGIN / START TRANSACTION
- COMMIT
- ROLLBACK

Example:

```
START TRANSACTION;
UPDATE accounts SET balance = balance - 100 WHERE id = 1;
COMMIT;
```

Question 5] What are window functions and how do you use them?

Answer] Window functions perform calculations across related rows without grouping.

Common functions: ROW_NUMBER(), RANK(), LAG(), LEAD().

Example:

```
SELECT name, salary,  
RANK() OVER (ORDER BY salary DESC) AS rank  
FROM employees;
```

Question 6] How do you optimize SQL queries?

Answer]

- Use indexes
 - Avoid SELECT *
 - Use proper joins
 - Analyze execution plans
 - Filter data early using WHERE
 - Use CTEs wisely
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Question 7] What are correlated subqueries?

Answer] A **correlated subquery** depends on values from the outer query and runs once per row.

Example:

```
SELECT name  
FROM employees e  
WHERE salary > (  
    SELECT AVG(salary) FROM employees WHERE dept = e.dept  
);
```

Question 8] What are Common Table Expressions (CTEs)?

Answer] CTEs are temporary named result sets used to simplify complex queries and improve readability.

Example:

```
WITH dept_avg AS (  
    SELECT dept, AVG(salary) avg_sal FROM employees GROUP BY dept  
)  
SELECT * FROM dept_avg;
```

Question 9] Explain data warehousing concepts in SQL.

Answer] Data warehousing focuses on analytical processing using:

- **Fact tables** (measurable data)
- **Dimension tables** (descriptive data)
- **Star & Snowflake schemas**
- ETL processes for data integration